LAB SHEET 3

Rebuilding with React (Components, Props, useState)

Course: Full Stack Web Development | **Department:** Computer Science & Engineering, COER University

Objective: Convert the product listing and cart into a proper React application using components, props, and state.

Tasks / Requirements:

1. Create a new Vite + React project: npm create vite@latest <yourstore>-react -- --template react, then npm install.
2. Move product data into a products.js file exporting an array of product objects acting as mock data.
3. Create a ProductCard component that receives a single product as props and renders its image, name, price, and an 'Add to Cart' button.
4. Create a ProductList component that imports products.js, maps over the array, and renders one ProductCard per product with a unique key.
5. In App.jsx, create cart state using useState. Pass an addToCart function down to ProductCard via props.
6. Create a Cart component that receives the cart array and a removeFromCart function as props, rendering items, quantities, subtotals, and grand total.
7. Add show/hide logic (using state) so the Cart can be toggled open/closed.
8. Create a CheckoutForm component with controlled inputs for Name, Address, Pincode, Phone, and Payment Method.
9. Add conditional rendering: if cart is empty, show 'Your cart is empty' message instead of table.

Solution -> products.js

```
export const products = [
  {
    id: "p1",
    name: "Wireless Headphones",
    price: 2999,
    image: "https://via.placeholder.com/150"
  },
  {
    id: "p2",
    name: "Smart Watch",
    price: 4999,
    image: "https://via.placeholder.com/150"
  },
  {
    id: "p3",
    name: "Gaming Mouse",
    price: 1299,
    image: "https://via.placeholder.com/150"
  }
];
```

ProductCard.jsx

```
export default function ProductCard({ product, onAddToCart }) {
  return (
    <div style={{ border: "1px solid #ccc", padding: "16px", borderRadius: "8px", width:
```

```
"200px" }}>
        <img src={product.image} alt={product.name} style={{ width: "100%", height: "auto" }} />
        <h3>{product.name}</h3>
        <p>₹{product.price}</p>
        <button onClick={() => onAddToCart(product)}>Add to Cart</button>
      </div>
  );
}
```

ProductList.jsx

```
import { products } from "../products";
import ProductCard from "./ProductCard";

export default function ProductList({ onAddToCart }) {
  return (
    <div style={{ display: "flex", gap: "16px", flexWrap: "wrap", padding: "20px 0" }}>
      {products.map((product) => (
        <ProductCard key={product.id} product={product} onAddToCart={onAddToCart} />
      ))}
    </div>
  );
}
```

Cart.jsx

```
export default function Cart({ cart, onRemoveFromCart }) {
  const grandTotal = cart.reduce((total, item) => total + item.product.price * item.quantity,
0);

  if (cart.length === 0) {
    return <p>Your cart is empty</p>;
  }

  return (
    <div style={{ border: "1px solid #aaa", padding: "16px", marginTop: "20px", borderRadius:
"8px" }}>
      <h2>Your Cart</h2>
      <table style={{ width: "100%", textAlign: "left" }}>
        <thead>
          <tr>
            <th>Item</th>
            <th>Price</th>
            <th>Quantity</th>
            <th>Subtotal</th>
            <th>Action</th>
          </tr>
        </thead>
        <tbody>
          {cart.map(({ product, quantity }) => (
            <tr key={product.id}>
              <td>{product.name}</td>
              <td>₹{product.price}</td>
              <td>{quantity}</td>
              <td>₹{product.price * quantity}</td>
              <td>
                <button onClick={() => onRemoveFromCart(product.id)}>Remove</button>
              </td>
            </tr>
          ))}
        </tbody>
```

```
        </table>
        <h3>Grand Total: ₹{grandTotal}</h3>
      </div>
  );
}
```

CheckoutForm.jsx

```
import { useState } from "react";

export default function CheckoutForm() {
  const [formData, setFormData] = useState({
    name: "",
    address: "",
    pincode: "",
    phone: "",
    paymentMethod: "COD"
  });

  const handleChange = (e) => {
    const { name, value } = e.target;
    setFormData((prev) => ({ ...prev, [name]: value }));
  };

  const handleSubmit = (e) => {
    e.preventDefault();
    console.log("Order Submitted:", formData);
    alert("Order placed successfully!");
  };

  return (
    <form onSubmit={handleSubmit} style={{ display: "flex", flexDirection: "column", gap:
"10px", maxWidth: "300px", marginTop: "20px" }}>
      <h2>Checkout Details</h2>
      <input type="text" name="name" placeholder="Name" value={formData.name}
onChange={handleChange} required />
      <input type="text" name="address" placeholder="Address" value={formData.address}
onChange={handleChange} required />
      <input type="text" name="pincode" placeholder="Pincode" value={formData.pincode}
onChange={handleChange} required />
      <input type="tel" name="phone" placeholder="Phone" value={formData.phone}
onChange={handleChange} required />
      <select name="paymentMethod" value={formData.paymentMethod} onChange={handleChange}>
        <option value="COD">Cash on Delivery</option>
        <option value="UPI">UPI</option>
        <option value="Card">Credit/Debit Card</option>
      </select>
      <button type="submit">Place Order</button>
    </form>
  );
}
```

App.jsx

```
import { useState } from "react";
import ProductList from "./components/ProductList";
import Cart from "./components/Cart";
import CheckoutForm from "./components/CheckoutForm";

export default function App() {
  const [cart, setCart] = useState([]);
```

```
  const [isCartOpen, setIsCartOpen] = useState(false);

  const handleAddToCart = (product) => {
    setCart((prevCart) => {
      const existingItem = prevCart.find((item) => item.product.id === product.id);
      if (existingItem) {
        return prevCart.map((item) =>
          item.product.id === product.id ? { ...item, quantity: item.quantity + 1 } : item
        );
      }
      return [...prevCart, { product, quantity: 1 }];
    });
  };

  const handleRemoveFromCart = (productId) => {
    setCart((prevCart) => prevCart.filter((item) => item.product.id !== productId));
  };

  const totalItemCount = cart.reduce((total, item) => total + item.quantity, 0);

  return (
    <div style={{ padding: "20px" }}>
      <nav style={{ display: "flex", justifyContent: "space-between", alignItems: "center",
borderBottom: "1px solid #ccc", paddingBottom: "10px" }}>
        <h1>My Store</h1>
        <button onClick={() => setIsCartOpen(!isCartOpen)}>
          Cart <span style={{ background: "red", color: "white", borderRadius: "50%",
padding: "2px 6px" }}>{totalItemCount}</span>
        </button>
      </nav>

      <ProductList onAddToCart={handleAddToCart} />

      {isCartOpen && (
        <>
          <Cart cart={cart} onRemoveFromCart={handleRemoveFromCart} />
          {cart.length > 0 && <CheckoutForm />}
        </>
      )}
    </div>
  );
}
```